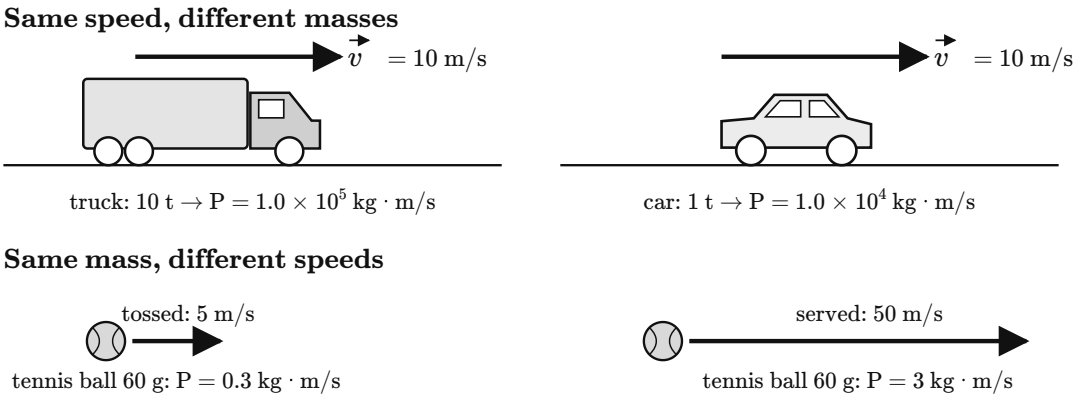
TUTORING CENTRE	Life Sciences · Physics Sheet **Linear Momentum** Grade 12 LS · Chapter 2 · Student toolkit	Lebanese curriculum

Name: Date:

1 Feel it | 2 Vector | 3 System | 4 Centre of mass | 5 Newton II | 6 Isolated system | 7 Projectile | Cards 1 · Practice 1 | 8 Collision & explosion | 9 Elastic collision | 10 Inelastic | Cards 2 · Practice 2

Part 1 — From intuition to Newton's second law

1 Feel it: who is harder to stop?



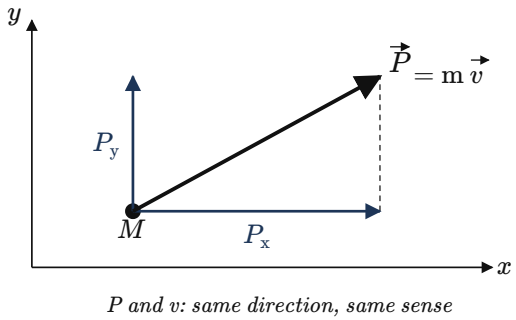
Stopping a moving body is hard when it is **heavy** or when it is **fast**. Catching a basketball is easy; catching a bowling ball at the same speed is not. A 10 g bullet is light, but at 400 m/s it goes through wood. A loaded oil tanker drifting slowly needs kilometres to stop. One quantity measures this "quantity of motion": the **linear momentum**.

2 The formula: a vector, not just a number

Definition A particle of mass m moving with the velocity $\vec{v}$ has the linear momentum

$$\vec{P} = m\,\vec{v} \qquad \text{(unit: kg·m/s)}$$

$\vec{P}$ has the **direction and sense of** $\vec{v}$; its magnitude is $P = mv$.



P and v: same direction, same sense

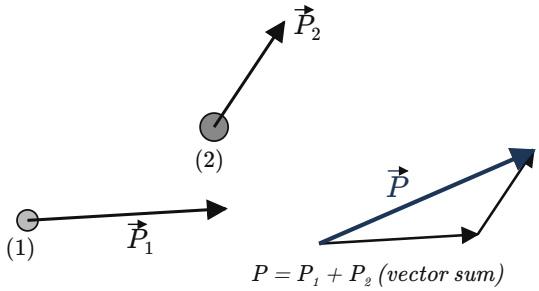
In a frame $(O, \vec{i}, \vec{j})$: $P_x = m\,v_x$ and $P_y = m\,v_y$.

Watch the sign. A ball of 0.2 kg moving at 4 m/s towards the **negative** sense of the axis: $\vec{v} = -4\,\vec{i}$, so $\vec{P} = -0.8\,\vec{i}$ (kg · m/s). The minus sign carries the direction.

3 Linear momentum of a system

System The linear momentum of a system is the **vector sum** of the momenta of its parts:

$$\vec{P}_{sys} = \vec{P}_1 + \vec{P}_2 + \cdots = m_1\vec{v}_1 + m_2\vec{v}_2 + \ldots$$

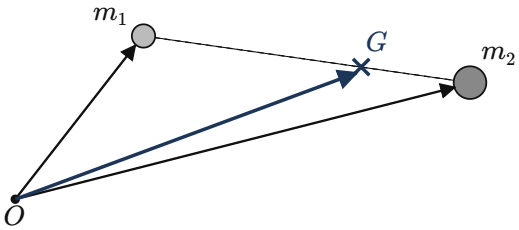


Surprise. Two carts move towards each other: $m_1 = 2$ kg, $\vec{v}_1 = 3\,\vec{i}$; $m_2 = 3$ kg, $\vec{v}_2 = -2\,\vec{i}$ (m/s).
$\vec{P}_{sys} = 6\,\vec{i} - 6\,\vec{i} = \vec{0}$: both carts move, yet the system has zero momentum.

4 Centre of mass: a second road to the same momentum

Centre of mass G (also called centre of inertia) of two particles:

$$(m_1 + m_2)\,\overrightarrow{OG} = m_1\,\overrightarrow{OG_1} + m_2\,\overrightarrow{OG_2}$$



Derive once with respect to time, then derive again:

position
$(m_1 + m_2)\overrightarrow{OG} = m_1\overrightarrow{OG_1} + m_2\overrightarrow{OG_2}$

$\frac{d}{dt}$ →

velocity
$M\vec{v}_G = m_1\vec{v}_1 + m_2\vec{v}_2$

$\frac{d}{dt}$ →

acceleration
$M\vec{a}_G = \dfrac{d\vec{P}_{sys}}{dt}$

Two approaches, one result $\vec{P}_{sys} = \underbrace{m_1\vec{v}_1 + m_2\vec{v}_2}_{\text{sum of the parts}} = \underbrace{M\,\vec{v}_G}_{\text{total mass} \times \text{velocity of G}}$, with $M = m_1 + m_2$.
The second derivative links $\vec{P}$ to the acceleration of G: this opens the door to Newton's second law.

5 Newton's second law

Law In a Galilean (inertial) frame, the sum of the **external** forces acting on a system equals the time-derivative of its linear momentum:

$$\Sigma \vec{F}_{ext} = \frac{d\vec{P}}{dt}$$

For one particle of constant mass: $\frac{d\vec{P}}{dt} = m\frac{d\vec{v}}{dt} = m\,\vec{a}$, so $\Sigma \vec{F}_{ext} = m\,\vec{a}$. Internal forces cancel in pairs and never appear.

Situation	What you write
$\vec{P}(t)$ is given as a function of time	**Derive**: $\Sigma \vec{F}_{ext} = \frac{d\vec{P}}{dt}$. Official exam 2021: $\vec{P} = (-0.9t + 0.4)\,\vec{i} \Rightarrow \Sigma \vec{F} = -0.9\,\vec{i}$ (N).
$\Sigma \vec{F}_{ext}$ is **constant**	$\vec{P}$ varies linearly with t, so the derivative becomes a ratio of differences: $\Delta \vec{P} = \vec{P}_f - \vec{P}_i = \Sigma \vec{F}_{ext}\,\Delta t$.
Very short shock (kick, bounce)	Average force: $\Sigma \vec{F}_{avg} = \frac{\Delta \vec{P}}{\Delta t} = \frac{\vec{P}_f - \vec{P}_i}{\Delta t}$ — subtract **vectors**, signs included.
$\Sigma \vec{F}_{ext} = \vec{0}$	$\Delta \vec{P} = \vec{0}$: $\vec{P}_f = \vec{P}_i$. The momentum is conserved (step 6).

6 Isolated system: momentum is conserved

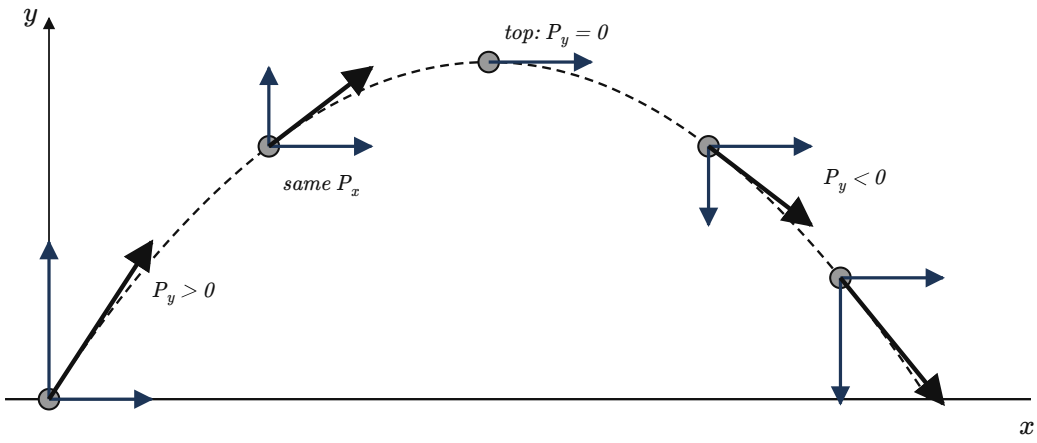
Definition A system is **isolated** if it is subject to no external force, or **pseudo-isolated** if the external forces cancel: $\Sigma \vec{F}_{ext} = \vec{0}$.
Then $\frac{d\vec{P}}{dt} = \vec{0}$: $\vec{P}_{sys} = \overrightarrow{\text{constant}}$, and G moves in a straight line at constant velocity (or stays at rest).

Isolated (pseudo-isolated)?	Why
Puck on an air table — **yes**	Weight and air-cushion reaction cancel; no friction.
Two skaters on smooth ice — **yes**	Weights and normal reactions cancel; friction neglected.
Ball in flight — **no**	The weight is not balanced.
Car braking on a road — **no**	The friction of the road is an external force.

Bonus — principle of interaction. For two bodies forming an isolated system: $\frac{d\vec{P}_1}{dt} + \frac{d\vec{P}_2}{dt} = \vec{0}$, so $\vec{F}_{2\to 1} = -\vec{F}_{1\to 2}$.

7 Parabolic motion: two components, two behaviours

A ball is thrown with $\vec{v}_0$ at the angle α above the horizontal; air resistance is neglected. The only force is the weight $m\vec{g} = -mg\,\vec{j}$.



Along x $\dfrac{dP_x}{dt} = 0$

$P_x = m\,v_0\cos\alpha = \text{constant}$

$v_x = v_0\cos\alpha$

Along y $\dfrac{dP_y}{dt} = -mg$

$P_y = -mg\,t + m\,v_0\sin\alpha$

$v_y = -g\,t + v_0\sin\alpha$

Reading the graphs (official-exam style). $P_x(t)$ is a horizontal line; $P_y(t)$ is a straight line of slope $-mg$, so $m = -\dfrac{\text{slope}}{g}$. At the top of the path $P_y = 0$, and $P = P_x$ is minimum.

✂ Cards 1 — cut, keep, revise

1 Linear momentum

$\vec{P} = m\vec{v}$ (kg · m/s). Same direction and sense as $\vec{v}$. Magnitude $P = mv$.

2 Components

$P_x = mv_x$, $P_y = mv_y$. Along one axis, the sign gives the sense: $\vec{P} = -0.8\,\vec{i}$ points backwards.

3 System

$\vec{P}_{sys} = \vec{P}_1 + \vec{P}_2 + \ldots$ — a **vector** sum. Moving parts can give $\vec{P}_{sys} = \vec{0}$.

4 Centre of mass

$M\overrightarrow{OG} = \Sigma m_i\overrightarrow{OG_i}$

derive: $\vec{P}_{sys} = M\vec{v}_G$

derive again: $\frac{d\vec{P}}{dt} = M\vec{a}_G$

5 Newton's second law

$\Sigma\vec{F}_{ext} = \dfrac{d\vec{P}}{dt}$. Only **external** forces. One particle: $\Sigma\vec{F} = m\vec{a}$.

6 Derivative or difference?

$\vec{P}(t)$ given → derive. Constant force → $\Delta\vec{P} = \Sigma\vec{F}\,\Delta t$. Shock → $\vec{F}_{avg} = \Delta\vec{P}/\Delta t$.

7 Isolated system

$\Sigma \vec{F}_{ext} = \vec{0} \Rightarrow \vec{P}_{sys}$ constant; G at rest or in uniform rectilinear motion.

8 Projectile

Only the weight: P_x constant; $P_y = -mgt + P_{0y}$. Top: $P_y = 0$. Slope of $P_y(t) = -mg$.

Practice 1 — after Part 1

≈ 1 hour · take $g = 10\ \text{m/s}^2$

Exercise 1 — Two pucks on an air table (6 pts)

Two pucks (A) and (B), of masses $m_A = 0.2$ kg and $m_B = 0.3$ kg, slide on a horizontal air table along the axis $(O, \vec{i})$. Their abscissas are: $x_A = 0.5\,t - 0.2$ and $x_B = -0.2\,t + 0.6$ (SI).

1.1 Determine the velocities $\vec{v}_A$ and $\vec{v}_B$. In which sense does each puck move?

1.2 Calculate $\vec{P}_A$, $\vec{P}_B$ and the linear momentum $\vec{P}$ of the system (A, B).

1.3 Show that the abscissa of the centre of mass G is $x_G = 0.08\,t + 0.28$ (SI). Deduce $\vec{v}_G$, then verify that $\vec{P} = (m_A + m_B)\,\vec{v}_G$.

1.4 Derive $\vec{P}$ with respect to time. What do you conclude about the external forces? Name them and explain why they cancel.

1.5 Specify the nature of the motion of G.

Exercise 2 — A sled in Mzaar, then a volleyball smash (7 pts)

Part A. A sled carrying a child (total mass $M = 40$ kg) slides down a straight snowy slope inclined by α with the horizontal ($\sin\alpha = 0.10$). The axis $(O, \vec{i})$ is along the slope, directed downward. The snow exerts a friction force $\vec{f}$ of constant magnitude. A phone app gives the linear momentum of the sled: $\vec{P} = (20\,t + 40)\,\vec{i}$ (SI).

A.1 Name the external forces acting on the system (sled + child).

A.2 Applying Newton's second law, determine $\Sigma \vec{F}_{ext}$. Deduce the value of f.

A.3 Determine the velocity of the sled at $t = 0$ and its acceleration. Specify the nature of its motion.

A.4 Calculate $\Delta\vec{P}$ between $t = 0$ and $t = 4$ s. Verify that $\Delta\vec{P} = \Sigma\vec{F}_{ext}\,\Delta t$. Why is this relation allowed here?

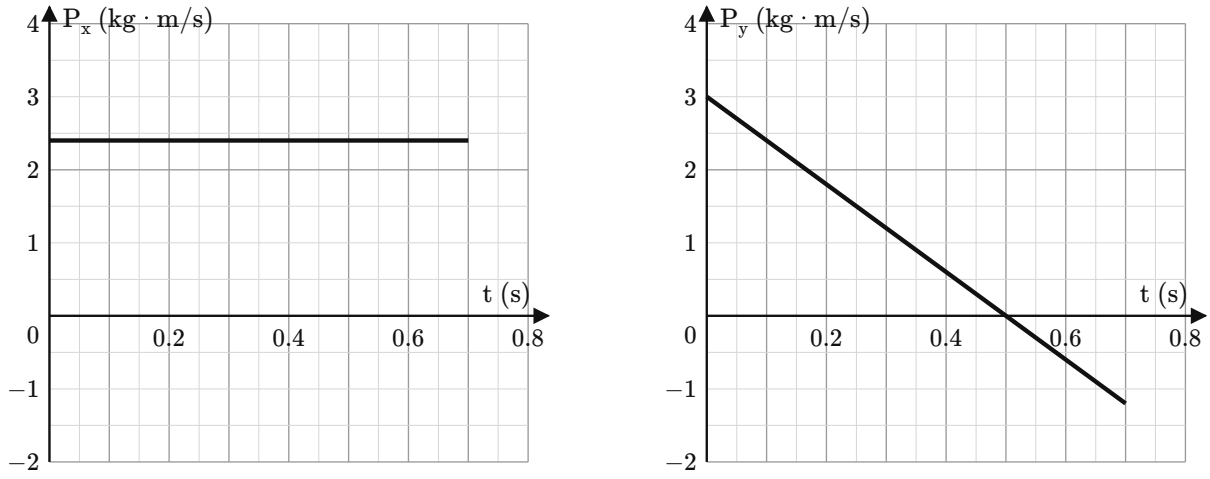
Part B. A volleyball ($m = 0.27$ kg) arrives horizontally at a player's hands at 8 m/s and leaves in the opposite sense at 12 m/s. The contact lasts $\Delta t = 0.02$ s. Take $\vec{i}$ in the sense of the final velocity.

B.1 Write $\vec{P}_i$ and $\vec{P}_f$. Calculate $\Delta\vec{P}$. (A student writes $\Delta P = 0.27 \times (12 - 8)$: find the mistake.)

B.2 Calculate the average force exerted by the hands on the ball. Compare it with the weight of the ball and conclude.

Exercise 3 — A basketball shot (7 pts)

A player throws a basketball from the height $h_0 = 2.0$ m at $t_0 = 0$. Air resistance is neglected. A video analysis gives the components of the linear momentum of the ball in the frame $(O, \vec{i}, \vec{j})$, $\vec{j}$ upward (Doc. 1). The ball enters the basket at $t = 0.7$ s.



Graph (a) **Graph (b)**

Doc. 1

3.1 Name the force acting on the ball during its flight. Is the ball an isolated system? Justify.

3.2 Applying Newton's second law, show that P_x is constant and that $\dfrac{dP_y}{dt} = -mg$. Are graphs (a) and (b) in agreement?

3.3 Using graph (b), determine the mass m of the ball.

3.4 Determine the components of $\vec{v}_0$, its magnitude, and the angle α of $\vec{v}_0$ with the horizontal ($\cos 51^\circ \approx 0.63$).

3.5 At which instant does the ball reach the top of its path? Justify using Doc. 1. What is then its linear momentum?

3.6 Determine $\vec{P}$ and the speed of the ball when it enters the basket ($h = 3.05$ m). Check this speed by an energetic method (Chapter 1).

Part 2 — Collisions and explosions

8 Why momentum is conserved in a collision or an explosion

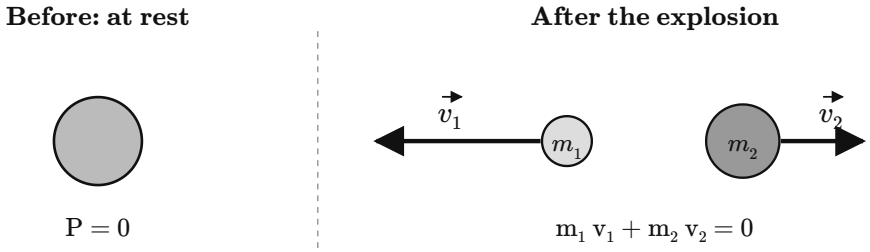
Key idea A collision or an explosion lasts a very short time. The internal forces between the bodies are huge; the external forces (weights, reactions, friction) are finite and act during a negligible time. During the interaction, the system is **pseudo-isolated**:

$$\vec{P}_{sys}(\text{just before}) = \vec{P}_{sys}(\text{just after})$$

Method (4 steps)
① System = **all** the bodies that interact. ② Choose an axis. ③ Write $\vec{P}$ just before and just after. ④ Equate, then project on the axis (signs!).

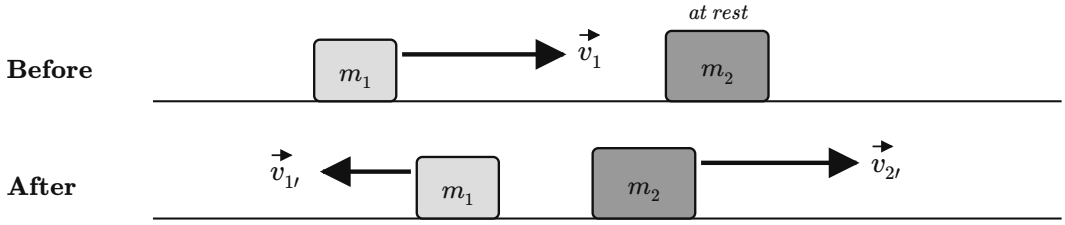
Trap. A ball bouncing on a wall: the wall (Earth) is an external body; the momentum of the **ball alone** is not conserved.

Explosion — a body at rest breaks into two parts (fireworks, rifle recoil, two skaters pushing each other).



Explosion from rest $\vec{0} = m_1\vec{v}_1 + m_2\vec{v}_2 \Rightarrow \vec{v}_2 = -\dfrac{m_1}{m_2}\,\vec{v}_1$: opposite senses, and the lighter part is the faster. G stays at rest. Kinetic energy appears: it comes from internal (chemical or elastic) energy.

9 Elastic collision — the main case



Elastic Both the linear momentum **and** the kinetic energy of the system are conserved. Head-on collision, m_2 initially at rest:

$$m_1 v_1 = m_1 v_1' + m_2 v_2' \qquad \tfrac{1}{2}m_1 v_1^2 = \tfrac{1}{2}m_1 v_1'^2 + \tfrac{1}{2}m_2 v_2'^2$$

Solving trick. Rewrite: $m_1(v_1 - v_1') = m_2 v_2'$ and $m_1(v_1^2 - v_1'^2) = m_2 v_2'^2$. Divide: $v_1 + v_1' = v_2'$. Then:

$$v_1' = \frac{m_1 - m_2}{m_1 + m_2}\,v_1 \qquad v_2' = \frac{2m_1}{m_1 + m_2}\,v_1$$

Case	Result
$m_1 = m_2$	Exchange: $v_1' = 0$, $v_2' = v_1$ (billiards, Newton's cradle)
$m_1 < m_2$	$v_1' < 0$: (1) bounces back
$m_1 > m_2$	Both move forward

"Show that the collision is elastic" (official exam 2021): calculate the kinetic energy of the system just before and just after, and show they are equal.

10 Inelastic collision — just know it

Momentum is conserved, but kinetic energy is **not**: part of it becomes internal energy (deformation, heat, sound). If the bodies **stick together** (perfectly inelastic or "soft" collision), they share a common velocity: $\vec{V} = \dfrac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$.

✂ Cards 2 — cut, keep, revise

9 When is $\vec{P}$ conserved?
During a collision or an explosion (very short): system pseudo-isolated $\rightarrow \vec{P}_{before} = \vec{P}_{after}$.

10 Method
System = all interacting bodies · axis · $\vec{P}$ just before = $\vec{P}$ just after · project with signs.

11 Explosion from rest
$m_1\vec{v}_1 + m_2\vec{v}_2 = \vec{0}$: opposite senses; $\frac{v_1}{v_2} = \frac{m_2}{m_1}$. KE comes from internal energy.

12 Elastic collision
$\vec{P}$ conserved **and** KE conserved. Check: $KE_{before} = KE_{after}$.

13 Head-on, target at rest
$v_1' = \frac{m_1 - m_2}{m_1 + m_2}v_1$, $v_2' = \frac{2m_1}{m_1 + m_2}v_1$. Equal masses: exchange of velocities.

14 Inelastic
$\vec{P}$ conserved, KE not. Stuck together: $\vec{V} = \frac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1 + m_2}$.

Practice 2 — after Part 2

≈ 1 hour · take $g = 10\ \text{m/s}^2$

Exercise 1 — Fireworks and a hunting rifle (6 pts)

Part A. A firework shell ($M = 1.2$ kg) is launched vertically. At the top of its path, where its velocity is zero, it explodes into two fragments: (1) of mass $m_1 = 0.4$ kg, thrown horizontally with $\vec{v}_1 = 30\,\vec{i}$ (m/s), and (2) of mass m_2.

A.1 Why can the shell be considered a pseudo-isolated system during the explosion, although it is subject to its weight?

A.2 What is the linear momentum of the shell just before the explosion? Determine $\vec{v}_2$.

A.3 Calculate the kinetic energy gained by the system. Where does it come from?

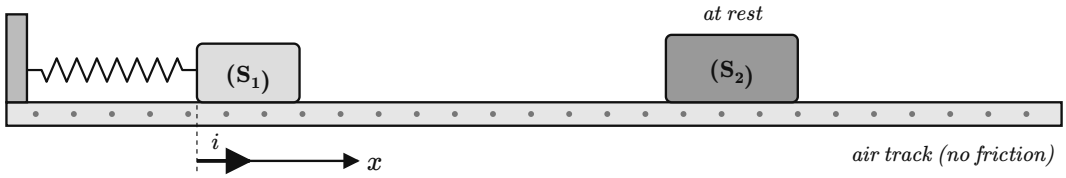
A.4 What is the velocity of the centre of mass G just after the explosion?

Part B. A hunting rifle (4.0 kg) fires a 10 g bullet at 400 m/s. The rifle is initially at rest.

B.1 Calculate the recoil velocity of the rifle held loosely.

B.2 The hunter (70 kg) holds the rifle firmly against his shoulder. Calculate the new recoil velocity and explain why holding it firmly is safer.

Exercise 2 — Elastic collision on an air track (8 pts)



Doc. 2

Glider (S_1): $m_1 = 0.2$ kg; glider (S_2): $m_2 = 0.3$ kg, initially at rest. Spring: stiffness $k = 80$ N/m, compressed by $d = 5$ cm, then released; (S_1) leaves the spring before hitting (S_2). Friction is negligible.

2.1 Using the conservation of mechanical energy, show that (S_1) leaves the spring with the speed $v_1 = 1.0$ m/s.

2.2 Why is the momentum of the system (S_1, S_2) conserved during the collision?

2.3 The collision is elastic. Write the two conservation relations in terms of v_1' and v_2' (algebraic values along $\vec{i}$).

2.4 Deduce that $v_1 + v_1' = v_2'$, then calculate v_1' and v_2'. Interpret the sign of v_1'.

2.5 Verify numerically the conservation of $\vec{P}$ and of the kinetic energy.

2.6 Calculate the velocity of G before and after the collision. Conclude.

2.7 (S_2) is replaced by a glider of mass 0.2 kg. Predict, without calculation, the velocities after an elastic collision.

2.8 *Just to know.* With Velcro, the two original gliders stick together. Calculate their common speed and the kinetic energy transformed into internal energy.

Exercise 3 — Spot the interaction (6 pts)

For each situation: give the type (elastic collision, perfectly inelastic collision, explosion, or none of them), say whether the momentum of the indicated system is conserved, and calculate the missing value.

(a) Billiards. The white ball (170 g) at 2.0 m/s hits head-on the red ball (170 g) at rest; the white ball stops. System: the two balls. Find the speed of the red ball, then check the type.

(b) Dart. A 20 g dart at 10 m/s sticks into a 180 g cork block resting on a frictionless cart (cart mass neglected). System: dart + block. Find their common speed and the kinetic energy lost.

(c) Ice rink. Two skaters (60 kg and 40 kg), at rest, push each other; the 40 kg skater leaves at 3.0 m/s. System: the two skaters. Find the velocity of the other skater.

(d) Wall. A tennis ball (60 g) hits a wall perpendicularly at 20 m/s and bounces back at 20 m/s. System: the ball alone. Calculate $\Delta\vec{P}$ of the ball. Is its momentum conserved? Its kinetic energy?